

Level: Bachelor
Programme: BE
Course: Calculus II

Semester: Fall

Year : 2023
Full Marks: 100
Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

b) Find the directional derivative of $f = x^2yz + 4xz^2$ at $(1, -2, 1)$ in the direction $2\vec{i} - \vec{j} - 2\vec{k}$. 5

c) Calculate: $\oint_C \vec{F} \cdot d\vec{r}$ if $\vec{F} = (y, z, x)$, $C: \vec{r} = (t, t^2, t^3)$, from $(0, 0, 0)$ and $(2, 4, 8)$. 5

5. a) Define Green's Theorem. Evaluate $\oint_C \vec{F} \cdot d\vec{r}$, where 8

$\vec{F} = (\sin y, \cos x)$ and c is the triangle with vertices $(0, 0), (\pi, 0), (\pi, 1)$, by using Green's theorem.

b) Evaluate $\oint_C \vec{F} \cdot d\vec{r}$ by using Stoke's theorem, where $\vec{F} = (y^2, z^2, x^2)$ 7

and c is the boundary of the surface $S: x + y + z = 1$ in the first octant.

OR

Evaluate the surface integral $\iint_S \vec{F} \cdot \vec{n} dA$ where $\vec{F} = (x^2, y^2, z^2)$ and $s: \vec{r} = (u \cos v, u \sin v, 3v)$, $0 \leq u \leq 1$, $0 \leq v \leq 2\pi$

6. a) Find the fourier series of $f(x) = x + |x|$ for $-\pi < x < \pi$. 7

b) Expand the function $f(x) = x^2$ in the interval $0 < x \leq \pi$ in half range Fourier cosine series and show that $\frac{\pi^2}{6} = 1 + \frac{1}{4} + \frac{1}{9} + \frac{1}{16} + \dots$ 8

7. Solve Any Two: 2x5

a) Find the general solution of $2u_x + 2u_y - u = 0$.

b) Derive the one-dimensional traffic flow model using conservation law.

c) Show that the value under integral sign $\int_{(4,0,3)}^{(-1,1,2)} [(yz + 1)dx + (xz + 1)dy + (xy + 1)dz]$ is exact and evaluate the integral.

1. a) Evaluate $\int_0^2 \int_0^{\sqrt{4-y^2}} \cos(x^2 + y^2) dx dy$ by changing into polar integration. 5

b) Evaluate: $\int_0^1 \int_0^1 \int_0^1 (x^2 + y^2 + z^2) dz dy dx$ 5

c) Find the volume in the first octant bounded by the coordinate planes, the cylinder $x^2 + y^2 = 4$ and $z + y = 3$. 5

2. a) Solve the differential equation $y'' + (1 - x^2)y = 0$ by using power series method. 7

b) Define Legendre's equation. Also derive the solution of Legendre's equation. 8

OR

If $J_n(x)$ represents the Bessel's function of order n then show that:

$$i. \frac{d}{dx} [x^n J_n(x)] = x^n J_{n-1}(x).$$

$$ii. J_{\frac{1}{2}}(x) = \sqrt{\frac{2}{\pi x}} \sin x,$$

3. a) State First Shifting Theorem of Laplace transforms. 8

i. Find the Laplace transform of $te^{-t} \cosh t$

ii. Find the inverse Laplace transform of $\frac{1}{s^2 - 5s + 6}$

b) Solve the initial value problem: $y'' + 4y' + 3y = e^{-t}$, $y(0) = y'(0) = 1$ by using Laplace transform. 7

4. a) If $\Phi = \log(x^2 + y^2 + z^2)$ then find $\text{div}(\text{grad} \Phi)$. 5